



Evaluating the Impact of Immersive Technologies
on
STEM Education in Sri Lanka's Secondary and
Tertiary Curriculum
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Feasibility Study

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1. Executive Summary

This feasibility study evaluates the development and implementation of a specialized Augmented Reality (AR) infrastructure designed to modernize Mathematics education for Grade 8 students within the Sri Lankan school system. While the national curriculum is well-structured, mathematics instruction currently relies heavily on static 2D illustrations and theoretical explanations to describe complex spatial and abstract concepts, which often leads to rote memorization rather than deep conceptual understanding. The primary objective of this component is to develop a robust mobile AR application that allows students to visualize abstract mathematical concepts—such as 3D geometry transformations, complex function graphs, and statistical data representations—by overlaying interactive 3D simulations directly onto their standard government-issued textbooks.

The proposed technical solution utilizes a smartphone camera to scan existing 2D textbook diagrams, triggering dynamic 3D models and animations through the integration of Unity and Vuforia's image-target recognition. By transforming flat diagrams of geometric shapes or algebraic patterns into rotatable, interactive simulations, the platform shifts the learning process from passive observation to active discovery. This approach is specifically designed for high accessibility in the local context, leveraging the mobile devices students already own and integrating seamlessly with physical textbooks without requiring expensive laboratory infrastructure or specialized equipment.

A detailed feasibility assessment confirms the technical viability and operational suitability of the system. Technical analysis proves that by using real-time 3D rendering optimization, high-quality educational content can be delivered effectively on mid-range Android devices, which is critical for widespread adoption in Sri Lankan classrooms and for functioning offline. Market evaluation highlights a significant research gap in localized AR tools specifically tailored for the Sri Lankan Grade 8 mathematics curriculum, indicating a strong demand for innovative digital pedagogy that supports spatial reasoning and problem-solving skills.

The positive findings of this study support the advancement of the Mathematics component as a cornerstone of the broader STEM learning module. By making fundamental mathematical principles tangible and accessible, the application has the potential to transform math education from a theoretical subject into an experiential one. This project addresses critical local challenges in resource-constrained environments while aligning with global trends in interactive media, positioning it as a scalable and impactful tool for modernizing mathematical literacy in Sri Lanka.

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2. Introduction

2.1 Background

Mathematics is a foundational subject, but many Grade 8 students find it difficult because of its abstract nature. In Sri Lanka, topics such as 3D shapes, geometric transformations, and function graphs are taught primarily through lecture-based instruction and static textbook illustrations. These traditional methods fail to support spatial reasoning, making it hard for students to visualize how shapes move or how variables change on a graph.

Augmented Reality (AR) can bridge this gap by overlaying interactive 3D simulations directly onto textbook diagrams. For example, a flat drawing of a cube can become a rotatable 3D object on a smartphone screen. This turns passive learning into an active discovery process, helping students grasp difficult concepts more quickly.

2.2 Objective

- **Visualize 3D Geometry:** Create interactive models for 3D shapes and transformations (rotations, reflections).
- **Interactive Algebra:** Develop AR tools to visualize graphing functions and data representation.
- **Improve Spatial Reasoning:** Design methodologies to evaluate how these 3D visuals help students solve problems.
- **Curriculum Alignment:** Ensure all models match the Sri Lankan Grade 8 Mathematics syllabus.

2.3 Scope

- **Technical:** Testing how well Unity and Vuforia handle mathematical models on mid-range Android phones.
- **Operational:** Making sure the app is easy for Grade 8 students to use with their existing textbooks.
- **Educational:** Focusing on the specific "unmet needs" of local students in geometry and algebra.

3. Problem Statement

STEM education in Sri Lanka, specifically in the field of Mathematics, faces major challenges in teaching abstract and spatial concepts through traditional textbooks. Current math lessons for Grade 8 students rely heavily on flat, two-dimensional diagrams and lecture-based instruction to explain complex geometric and algebraic systems. These conventional methods often fail to show how three-dimensional relationships work, such as the unfolding of a geometric solid into a net or the dynamic transformation of a shape through rotation and reflection. Because these concepts require spatial reasoning and the visualization of movement, static images cannot accurately demonstrate how they function in a real-world mathematical context.

Grade 8 serves as a critical turning point where mathematical topics become significantly more abstract, requiring students to interpret 3D relationships and dynamic interactions that are

difficult to visualize using only a blackboard. Current teaching methods lack interactive visualization and hands-on exploration, which limits a student's ability to connect theoretical formulas with practical understanding. As a result, students often resort to rote memorization to pass exams rather than gaining a deep conceptual mastery of the subject. Misconceptions frequently arise because students cannot observe or manipulate the abstract structures they are reading about, leading to reduced engagement and a lack of interest in advanced STEM fields.

There is a clear and urgent need for an interactive learning system that provides real-time, 3D visualization to make these "invisible" mathematical principles tangible and easier to explore. Augmented Reality (AR) technology can bridge this gap by overlaying interactive 3D simulations directly onto the diagrams in government-issued Grade 8 textbooks. By using a smartphone to scan a page, students can observe and interact with models of geometric solids, coordinate planes, and transformations in a safe and low-cost environment. The proposed AR platform offers an accessible way to modernize math education, helping students in Sri Lankan classrooms move from passive observation to active discovery and improved conceptual understanding.

4. Project Description

The proposed AR-based educational system uses real-time 3D simulations to create an interactive learning experience specifically for Grade 8 Mathematics students. As the Math Lead, this component focuses on using detailed 3D models that allow students to observe and manipulate abstract mathematical concepts that are traditionally difficult to visualize. These real-time interactions enable students to explore spatial relationships, such as how geometric solids are formed or how algebraic functions change, helping them understand the dynamic nature of mathematics.

A defining feature of the system is its ability to overlay interactive 3D content directly on textbook diagrams using mobile devices and Vuforia image-target recognition. Students can rotate models, manipulate parameters—such as changing the dimensions of a 3D object or shifting coordinates on a graph—and observe these simulations from multiple perspectives to gain conceptual clarity. This component specifically targets the Grade 8 curriculum, focusing on 3D geometry to improve spatial reasoning, geometric transformations where students interactively perform rotations and reflections, and algebraic functions to visualize how variables affect function graphs in a tangible 3D space.

Guided learning modules provide structured explanations aligned with the Sri Lankan national curriculum, while interactive feedback mechanisms support self-paced exploration and repeated practice. By integrating visualization and interactivity, the AR platform offers a risk-free and engaging approach to Mathematics. The system has the potential to transform classroom learning by making abstract mathematical abstractions tangible, increasing student

engagement, and supporting teachers in delivering effective STEM instruction in resource-constrained environments.

5. Feasibility Analysis

5.1 Technical Feasibility

The technical check shows that this project can be done using standard tools and midrange mobile phones.

- **Tools:** Use Unity for the app, Vuforia for scanning textbook markers, and Blender for 3D geometric models.
- **Accuracy:** Mathematical models must be precise to maintain educational credibility.
- **Performance:** Animations must stay smooth on the mid-range phones typically used by local students.

5.2 Market Feasibility

The market study shows a strong need for new way to teach Mathematics in Sri Lankan schools, specifically focusing on Grade 8 students.

- **Target:** Grade 8 students and math teachers in Sri Lanka.
- **Novelty:** There is a significant gap in MR/AR tools specifically designed for the Sinhala-language Grade 8 math curriculum.

5.3 Financial Feasibility

- **Budget:** Estimated cost is Rs. 20,000-30,000, covering software and basic hardware.
- **Usage:** The app will work offline, making it ideal for schools with limited internet access.

5.4 Operational Feasibility

- **Team Requirements** - We need a small team including AR developers to build the app, UI/UX designers to make it easy to use, and 3D animators to create the IT models.

- **Easy to Use** - The app is designed to work with existing school textbooks, so teachers do not need to change how they teach.
- **Works Anywhere** - The app can be used in both city and village schools because it works on regular mobile phones and does not need the internet to function.
- **Simple Training** - We will provide basic training for teachers and students, so they will feel comfortable using the AR tools during lessons.
- **Regular Updates** - We will use feedback from students to keep the app updated and make sure it stays helpful for a long time.

5.5 Environmental or Ethical Feasibility

- **Environmental Impact** - Because the AR app is fully digital, it has almost no negative impact on the environment. It allows students to explore IT concepts without needing physical lab setups or electronic waste from old hardware.
- **Sustainability** - The app replaces resource-heavy experiments with digital simulations, promoting a low-cost and eco-friendly way to learn about technology.
- **Ethical Standards** - All IT content is checked by educators to make sure it is accurate and follows the national curriculum.
- **Inclusivity and Privacy** - The system is designed to be inclusive for all students and strictly protects student data privacy to ensure a safe digital learning environment.

6. Risk Analysis

- **Device Compatibility** - The AR application must be optimized to run smoothly on a wide variety of Android devices, ranging from older models to modern smartphones common in Sri Lanka..
- **Software Stability and Performance** - It is critical to ensure the app is free of bugs and maintains high-quality, real-time 3D rendering of complex geometric processes without lagging.
- **Image Recognition Reliability** - Using Vuforia, the app must consistently recognize 2D math textbook diagrams under different lighting conditions to trigger the 3D models.
- **Technical Accuracy** Mathematical simulations, such as 3D geometry transformations and algebraic graphs, must be perfectly accurate to maintain educational credibility.

2. Financial Risks

- **Initial Development Costs** - Funding is required for specialized AR development, high-quality 3D modeling of mathematical objects, and software licensing fees.
- **Adoption Constraints** - Some schools may hesitate to adopt the technology due to limited budgets or a lack of familiarity with digital learning tools.

3. Operational Risks

- **User Training and Support** - Teachers and students may need initial guidance and training to feel confident using the AR tools during math lessons.
- **Maintenance and Updates** - The application will require regular technical support, troubleshooting, and updates to stay compatible with new mobile operating systems.
- **Scalability Challenges** - Expanding the math modules to include more advanced topics or additional grade levels will require more technical resources and expertise.

7. Recommendations

- **Work with Experts** - Collaborate with Grade 8 math teachers and curriculum experts early on to ensure the 3D geometric models and transformations are mathematically correct.
- **Build a Prototype** - Create a test version of the app to check how well it scans Grade 8 textbook pictures and to fix any technical bugs before the final release.
- **Get Student Feedback** - Test the app with students to see if the interface is easy to use and if it effectively helps them understand 3D geometry and algebra better.
- **Focus on Speed** - Ensure all 3D animations stay smooth and fast, even on older or cheaper mobile phones, to prevent student frustration during lessons.

8. Conclusion

The proposed Augmented Reality (AR) system for Grade 8 Mathematics education is a practical and innovative solution for schools in Sri Lanka. By using AR, the platform allows students to finally "see" and interact with difficult, abstract math concepts—like 3D geometry and coordinate graphs—that were previously limited to flat, static pictures in a textbook. This transition from passive reading to active 3D exploration improves student understanding and keeps them more engaged in the learning process.

The feasibility study confirms that this component is technically achievable using standard mobile devices and works effectively in local classrooms without needing the internet. It offers a low-cost way to modernize math education, making high-quality learning accessible to all students regardless of their school's physical facilities. Ultimately, this AR platform has the potential to transform traditional textbooks into an immersive experience, helping students build the spatial reasoning and digital skills they need for the future.

9. Reference

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